

Lesson 4 - Gradient Descent

⇒ Gradient descent is an algorithm to iteratively find minimum/maxima of a function.

⇒ Some functions are very hard to optimize analytically, hence the requirement for gradient descent.

eg $f(x) = e^x - \log x$

$$f'(x) = e^x - \frac{1}{x}$$

$$f'(x) = 0 \Rightarrow e^x = \frac{1}{x}$$

$$\boxed{x \cdot e^x = 1}$$

↓
how do we find an analytical solution?

⇒ One brute force algorithm could be to just try all numbers in a particular range in a step interval to take the closest one. (binary search too)

⇒ We can use the information about the slope at a point to tell us if the minima is to left/right of it.

⇒
+ve slope → move left
-ve slope → move right.

⇒ How much to left/right?

$$x_{\text{new}} = x_{\text{old}} - \text{slope}.$$

$$x_n = x_0 - f'(x_0)$$

⇒ Now slopes can be chaotic if there is a very large slope.

To mitigate this we add a multiplier to the expression

$$\Rightarrow x_n = x_0 - \underbrace{(0.01)}_{\downarrow} f'(x_0)$$

learning rate.

⇒ We can iterate this a lot of times. Every iteration gets us closer to the minima upto a certain point.

Gradient descent steps

- ① → Define a learning rate α
→ Choose a starting point x_0 .
- ② → Update $x_k = x_{k-1} - \alpha f'(x_{k-1})$
- ③ → Repeat step 2 enough times to get close enough to true minimum.

Learning rate

- ⇒ The performance and accuracy of gradient descent will be highly influenced by learning rate.
- ⇒ Too big → might miss the minimum
Too small → might take forever.